

*The SAS System**The CORR Procedure*

8	price	age	bred	angus	frame
Variables:	weight	conditioned	registered		

Dependent Variable:

Y=Price in thousands of dollars

Independent variables:

X1: Age

X2: Bred

X3: Angus

X4: frame

X5: weight

X6: conditioned

X7: registered

Problem Statement:

Determining the factors that drive market prices is fundamental to agricultural economics and livestock management. This report presents a regression analysis of cattle prices using a dataset of 115 livestock sales conducted by Wapello Livestock Sales in Wapello, Iowa, from mid-September 1999 to the end of January 2000. The data, originally compiled for a 2001 student project by Jay Heindel at the University of Iowa, includes the final selling price of each cow along with key animal characteristics.

The objective of this analysis is to quantify the relationship between selling price and several explanatory variables, including age, weight, frame size, breeding status, registration, physical condition, and Angus. Using multiple regression techniques, we aim to identify which factors significantly influence price and to what extent. Understanding these relationships can provide insights for both buyers and sellers in the livestock market and demonstrate the application of regression methods to real-world economic data.

*The SAS System**The CORR Procedure*

Pearson Correlation Coefficients, N = 115 Prob > r under H0: Rho=0								
	price	age	bred	angus	frame	weight	conditioned	registered
price	1.0000 0	- 0.5266 1 <.0001	0.0999 9 0.2877	0.1771 1 0.0583	0.2509 5 0.0068	0.1456 9 0.1203	0.29839 0.0012	0.23626 0.0110
age	- 0.5266 1 <.0001	1.0000 0	0.1444 9 0.1234	- 0.1870 0 0.0454	- 0.1497 0 0.1103	0.1909 8 0.0409	-0.13418 0.1528	- 0.28284 0.0022
bred	0.0999 9 0.2877	0.1444 9 0.1234	1.0000 0	0.0298 3 0.7516	0.1607 8 0.0861	0.2396 6 0.0099	0.24629 0.0080	- 0.07036 0.4550
angus	0.1771 1 0.0583	- 0.1870 0 0.0454	0.0298 3 0.7516	1.0000 0	0.0615 4 0.5135	- 0.0714 9 0.4477	0.20229 0.0302	0.04082 0.6649
frame	0.2509 5 0.0068	- 0.1497 0 0.1103	0.1607 8 0.0861	0.0615 4 0.5135	1.0000 0	0.2024 5 0.0300	0.06250 0.5070	0.10938 0.2445
weight	0.1456 9 0.1203	0.1909 8 0.0409	0.2396 6 0.0099	- 0.0714 9 0.4477	0.2024 5 0.0300	1.0000 0	0.44729 <.0001	- 0.07854 0.4041
conditioned	0.2983 9 0.0012	- 0.1341 8 0.1528	0.2462 9 0.0080	0.2022 9 0.0302	0.0625 0 0.5070	0.4472 9 <.0001	1.00000	- 0.01675 0.8589
registered	0.2362 6 0.0110	- 0.2828 4 0.0022	- 0.0703 6 0.4550	0.0408 2 0.6649	0.1093 8 0.2445	- 0.0785 4 0.4041	-0.01675 0.8589	1.00000

Interpretations:**Correlation between the dependent and independent variables:**

There is a negative moderate correlation between price and age. There is a weak positive correlation between price and bred, angus, frame, weight, conditioned and registered variables.

Correlation between independent variables:

There is low correlation between the independent variables, which suggests that we don't have issues with multicollinearity.

*The SAS System**The REG Procedure**Model: MODEL1**Dependent Variable: price*

Number of Observations Read	115
Number of Observations Used	115

Analysis of Variance					
Source	DF	Sum of Squares	Mean Square	F Value	Pr > F
Model	7	712671	101810	9.84	<.0001
Error	107	1107433	10350		
Corrected Total	114	1820104			

Model:

Price = 883.10486 -38.71073X1 + 25.33973X2 + 14.32634X3 + 27.47494X4 + 24.16738X5 + 33.62339X6 + 27.517561X7

Testing for the overall model:

H0: B1=B2=B3=B4=B5=B6=B7= 0

H1: At least one of the Betas is significant or different from zero.

P-value is less than alpha = 0.05, therefore we reject the null hypothesis and conclude that there is a linear relationship between price and at least one of the independent variables(age, bred, angus, frame, weight, conditioned and registered)

Root MSE	101.73416	R-Square	0.3916
Dependent Mean	1061.30435	Adj R-Sq	0.3518
Coeff Var	9.58577		

R-squared

About 39.16% of the total variation in price is explained by the model:

Root MSE

On average, the deviation between the response variable and the predicted value is around 101.734116

Dependent Mean

The average value of the dependent variable price is 1061.30435

*The SAS System**The REG Procedure**Model: MODEL1**Dependent Variable: price*

Parameter Estimates					
Variable	DF	Parameter Estimate	Standard Error	t Value	Pr > t
Intercept	1	883.10486	148.95319	5.93	<.0001
age	1	-38.71073	6.67524	-5.80	<.0001
bred	1	25.33973	22.06441	1.15	0.2533
angus	1	14.32634	20.07339	0.71	0.4770
frame	1	27.47494	20.37304	1.35	0.1803
weight	1	24.16738	14.53960	1.66	0.0994
conditioned	1	33.62339	24.90344	1.35	0.1798
registered	1	27.51761	21.27741	1.29	0.1987

Testing for the contribution of the individual:**FOR AGE:**

H0: There is no linear relationship between price and age

H1: There is a linear relationship between price and age

Since P-value is less than alpha, then there is a linear relationship between price and age

FOR BRED:

H0: There is no linear relationship between price and bred

H1: There is a linear relationship between price and bred

Since P-value=0.2533 is greater than alpha=0.05, then there is no linear relationship between price and bred

FOR ANGUS:

H0: There is no linear relationship between price and angus

H1: There is a linear relationship between price and angus

*The SAS System**The REG Procedure**Model: MODEL1**Dependent Variable: price*

Since P-value=0.477 is greater than $\alpha=0.05$, then there is no linear relationship between price and angus

FOR FRAME:

H0: There is no linear relationship between price and frame

H1: There is a linear relationship between price and frame

Since P-value=0.1803 is greater than $\alpha=0.05$, then there is no linear relationship between price and frame

FOR WEIGHT:

H0: There is no linear relationship between price and weight

H1: There is a linear relationship between price and weight

Since P-value=0.0994 is greater than $\alpha=0.05$, then there is no linear relationship between price and weight

FOR CONDITION:

H0: There is no linear relationship between price and conditioned

H1: There is a linear relationship between price and conditioned

Since P-value=0.1798 is greater than $\alpha=0.05$, then there is no linear relationship between price and conditioned

FOR REGISTERED:

H0: There is no linear relationship between price and registered

H1: There is a linear relationship between price and registered

Since P-value=0.1987 is greater than $\alpha=0.05$, then there is no linear relationship between price and registered

Interpretation of the independent variables:

Holding constant the effect of bred, angus, frame, weight, conditioned and registered, when the age of the cow is increase by 1 year we expect the price of the cow to decrease by 38.7103 thousands of dollars.

Holding constant the effect of age, angus, frame, weight, conditioned and registered, when the cow is bred we expect the price to increase by 25.33973 thousands of dollars compared to when the cow is not bred.

*The SAS System**The REG Procedure**Model: MODEL1**Dependent Variable: price*

Holding constant the effect of age, bred, frame, weight, conditioned and registered, when the cow is angus we expect the price to increase by 14.32634 thousands of dollars compared to when it is not angus.

Holding constant the effect of age, bred, angus, weight, conditioned and registered, when the frame size of the cow is large we expect the price to increase by 27.47494 thousands of dollars compared to when it is not.

Holding constant the effect of age, bred, angus, frame, conditioned and registered, when the weight of the cow increases by 100lb we expect the price to increase by 24.16738 thousands of dollars.

Holding constant the effect of age, bred, angus, frame, weight and registered, when the cow is in good condition we expect the price to increase by 33.62339 thousands of dollars compared to when it is not in good condition.

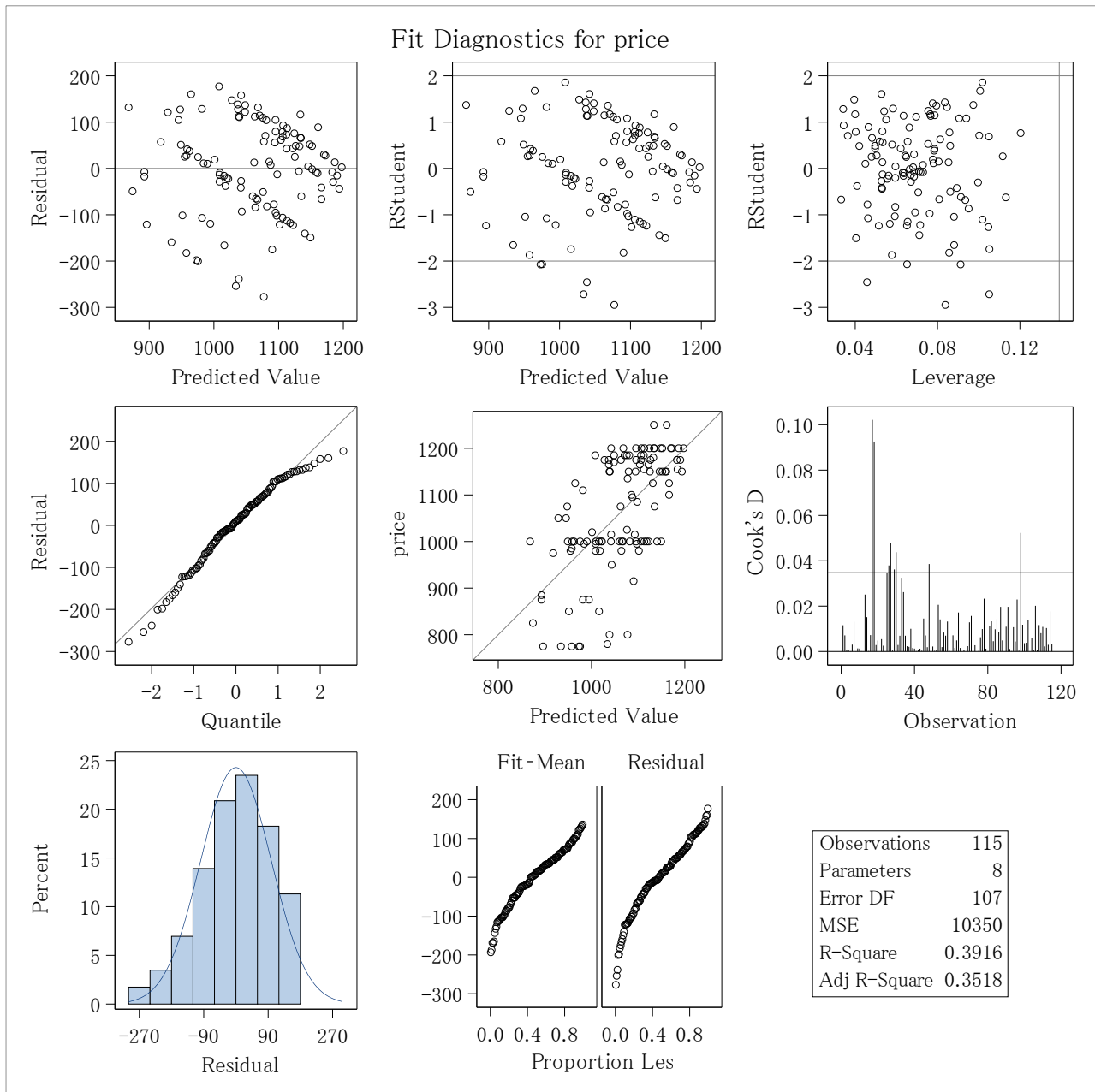
Holding constant the effect age, bred, angus, frame, weight and conditioned, when the cow is registered we expect the price of the cow to increase by 27.51761 thousands of dollars compared to when it is not registered.

The SAS System

The REG Procedure

Model: MODEL1

Dependent Variable: price



Interpretation of plots of the overall model:

Residual vs Predicted value: The points are randomly scattered around zero with no clear pattern which suggests that the assumptions of linearity is met and that there is evidence of constant variance.

RStudent vs Predicted value: It depicts no points outside three standard deviation from the mean therefore there is no outlier in the data.

Percent vs Residual; The histogram depicts a bell shaped(symmetrical) curve which indicated that the errors are normally distributed.

*The SAS System**The REG Procedure**Model: MODEL1**Dependent Variable: price*

Number of Observations Read	11 5
Number of Observations Used	11 5

Backward Elimination:
Step 0

All Variables Entered: R-Square = 0.3916 and C(p) = 8.0000

Analysis of Variance					
Source	D F	Sum of Squares	Mean Square	F Value	Pr > F
Model	7	712671	101810	9.84	<.0001
Error	107	1107433	10350		
Corrected Total	114	1820104			

*The SAS System**The REG Procedure**Model: MODEL1**Dependent Variable: price**Backward Elimination: Step 0*

Variable	Parameter Estimate	Standard Error	Type II SS	F Value	Pr > F
Intercept	883.10486	148.95319	363797	35.15	<.0001
age	-38.71073	6.67524	348067	33.63	<.0001
bred	25.33973	22.06441	13651	1.32	0.2533
angus	14.32634	20.07339	5271.84780	0.51	0.4770
frame	27.47494	20.37304	18823	1.82	0.1803
weight	24.16738	14.53960	28595	2.76	0.0994
conditioned	33.62339	24.90344	18867	1.82	0.1798
registered	27.51761	21.27741	17311	1.67	0.1987

**Bounds on condition number: 1.4918,
61.192**

**Backward Elimination:
Step 1**

**Variable angus Removed: R-Square = 0.3887 and C(p) =
6.5094**

*The SAS System**The REG Procedure**Model: MODEL1**Dependent Variable: price**Backward Elimination: Step 1*

Analysis of Variance					
Source	DF	Sum of Squares	Mean Square	F Value	Pr > F
Model	6	707400	117900	11.44	<.0001
Error	108	1112705	10303		
Corrected Total	114	1820104			

Variable	Parameter Estimate	Standard Error	Type II SS	F Value	Pr > F
Intercept	908.10554	144.44635	407207	39.52	<.0001
age	-39.19859	6.62505	360677	35.01	<.0001
bred	25.56709	22.01194	13900	1.35	0.2480
frame	28.39598	20.28589	20187	1.96	0.1644
weight	22.52168	14.32294	25474	2.47	0.1188
conditioned	37.63961	24.20415	24915	2.42	0.1229
registered	27.41621	21.22855	17184	1.67	0.1993

**Bounds on condition number: 1.4543,
45.059**

**Backward Elimination:
Step 2**

**Variable bred Removed: R-Square = 0.3810 and C(p) =
5.8523**

*The SAS System**The REG Procedure**Model: MODEL1**Dependent Variable: price**Backward Elimination: Step 2*

Analysis of Variance					
Source	DF	Sum of Squares	Mean Square	F Value	Pr > F
Model	5	693500	138700	13.42	<.0001
Error	109	1126604	10336		
Corrected Total	114	1820104			

Variable	Parameter Estimate	Standard Error	Type II SS	F Value	Pr > F
Intercept	904.04739	144.63515	403812	39.07	<.0001
age	-37.89780	6.54015	347054	33.58	<.0001
frame	32.25354	20.04420	26762	2.59	0.1105
weight	23.52362	14.31981	27892	2.70	0.1033
conditioned	43.45543	23.71847	34694	3.36	0.0697
registered	26.69955	21.25354	16311	1.58	0.2117

**Bounds on condition number: 1.449,
31.158**

**Backward Elimination:
Step 3**

**Variable registered Removed: R-Square = 0.3721 and C(p) =
5.4283**

*The SAS System**The REG Procedure**Model: MODEL1**Dependent Variable: price**Backward Elimination: Step 3*

Analysis of Variance					
Source	DF	Sum of Squares	Mean Square	F Value	Pr > F
Model	4	677189	169297	16.29	<.0001
Error	110	1142916	10390		
Corrected Total	114	1820104			

Variable	Parameter Estimate	Standard Error	Type II SS	F Value	Pr > F
Intercept	924.71894	144.07319	428030	41.20	<.0001
age	-40.01859	6.33510	414609	39.90	<.0001
frame	34.14675	20.03992	30167	2.90	0.0912
weight	23.22518	14.35542	27196	2.62	0.1086
conditioned	42.11669	23.75670	32656	3.14	0.0790

**Bounds on condition number: 1.4486,
20.171**

**Backward Elimination:
Step 4**

**Variable weight Removed: R-Square = 0.3571 and C(p) =
6.0560**

*The SAS System**The REG Procedure**Model: MODEL1**Dependent Variable: price**Backward Elimination: Step 4*

Analysis of Variance					
Source	DF	Sum of Squares	Mean Square	F Value	Pr > F
Model	3	649992	216664	20.55	<.0001
Error	111	1170112	10542		
Corrected Total	114	1820104			

Variable	Parameter Estimate	Standard Error	Type II SS	F Value	Pr > F
Intercept	1151.51997	33.48615	12465710	1182.53	<.0001
age	-36.73334	6.04440	389330	36.93	<.0001
frame	42.20588	19.55186	49122	4.66	0.0330
conditioned	60.96599	20.85397	90095	8.55	0.0042

**Bounds on condition number: 1.0396,
9.2539**

**All variables left in the model are significant at the
0.0500 level.**

*The SAS System**The REG Procedure**Model: MODEL1**Dependent Variable: price*

Summary of Backward Elimination							
Step	Variable Removed	Number Vars In	Partial R-Square	Model R-Square	C(p)	F Value	Pr > F
1	angus	6	0.0029	0.3887	6.5094	0.51	0.4770
2	bred	5	0.0076	0.3810	5.8523	1.35	0.2480
3	registered	4	0.0090	0.3721	5.4283	1.58	0.2117
4	weight	3	0.0149	0.3571	6.0560	2.62	0.1086

*The SAS System**The REG Procedure**Model: MODEL1**Dependent Variable: price*

Number of Observations Read	115
Number of Observations Used	115

Analysis of Variance					
Source	DF	Sum of Squares	Mean Square	F Value	Pr > F
Model	3	649992	216664	20.55	<.0001
Error	111	1170112	10542		
Lack of Fit	20	182865	9143.23525	0.84	0.6566
Pure Error	91	987247	10849		
Corrected Total	114	1820104			

Testing the ANOVA for the reduced model

H0: There is no linear relationship between age, frame, conditioned and price

H1: There is a linear relationship between price and at least one of the three independent variables

P-value is less than $\alpha = 0.05$, we reject the null hypothesis and conclude that there is a linear relationship between price and at least one of the three independent variables

Interpretation of lack of fit

H0: There is no lack of fit

H1: There is a lack of fit

P-value is greater than the significant level, therefore we fail to reject the null hypothesis and conclude that there is no lack of fit which means the model is adequate for the data.

Root MSE	102.67204	R-Square	0.3571
Dependent Mean	1061.30435	Adj R-Sq	0.3397
Coeff Var	9.67414		

R-squared:

About 35,71% of the total variation in price is explained by the model

Root MSE:

*The SAS System**The REG Procedure**Model: MODEL1**Dependent Variable: price*

On average, the deviation between the response variable and the predicted value is around 102.67204

Dependent Mean:

The average value of the dependent variable price is 1061.30435 thousands of dollars

Parameter Estimates								
Variable	DF	Parameter Estimate	Standard Error	t Value	Pr > t	Variance Inflation	95% Confidence Limits	
Intercept	1	1151.51997	33.48615	34.39	<.0001	0	1085.16494	1217.87500
age	1	-36.73334	6.04440	-6.08	<.0001	1.03956	-48.71073	-24.75595
frame	1	42.20588	19.55186	2.16	0.0330	1.02484	3.46256	80.94920
conditioned	1	60.96599	20.85397	2.92	0.0042	1.02025	19.64246	102.28952

Interpretation of the slopes of the final model:

Reduced model : Price = 1151.51997 – 36.73334age + 42.20588frame + 60.96599conditioned

Interpretation of B1.

Holding constant the effect of all the independent variables, when we increase the age of the animal by one year we expect the price to decrease by 36.73334 dollars.

Interpretation of B2

Holding constant the effect of all the other independent variables, when frame size is large we expect the price to increase by 42.20588 dollars compared to when it is small.

Interpretation of B6

Holding constant the effect of all the independent variables, when the cow is in good condition we expect the price to increase by 60.96599 dollars compared to when it is not in good condition.

VIF

The variance inflation factor for all the independent variables is less than 10, therefore multicollinearity exists but it does not create problems.

*The SAS System**The REG Procedure**Model: MODEL1**Dependent Variable: price*

Durbin-Watson D	1.433
Number of Observations	115
1st Order Autocorrelation	0.272

Test for Serial Autocorrelation:

H0: There is no serial autocorrelation

H1: There is serial autocorrelation

Since the DW=1.433 is between 0 and 1.5 we reject the null hypothesis and conclude that there is a strong positive autocorrelation among the errors

*The SAS System**The REG Procedure**Model: MODEL1**Dependent Variable: price*

Output Statistics							
Obs	Dependent Variable	Predicted Value	Std Error Mean Predict	Residual	Std Error Residual	Student Residual	Cook's D
1	1000	1144	16.1917	-144.4918	101.4	-1.425	0.013
2	1250	1144	16.1917	105.5082	101.4	1.041	0.007
3	980	1029	15.3941	-48.8193	101.5	-0.481	0.001
4	1015	1066	14.5648	-50.5526	101.6	-0.497	0.001
5	995	1010	21.4489	-15.0591	100.4	-0.150	0.000
6	825	894.3866	23.6754	-69.3866	99.905	-0.695	0.007
7	850	931.1199	20.5288	-81.1199	100.6	-0.806	0.007
8	1150	1139	19.5360	10.9807	100.8	0.109	0.000
9	1150	1181	18.7001	-31.2252	101.0	-0.309	0.001
10	1200	1144	16.1917	55.5082	101.4	0.547	0.002
11	1200	1181	18.7001	18.7748	101.0	0.186	0.000
12	1000	992.0859	18.3001	7.9141	101.0	0.078	0.000
13	1000	894.3866	23.6754	105.6134	99.905	1.057	0.016
14	1050	931.1199	20.5288	118.8801	100.6	1.182	0.015
15	1075	1047	20.8165	28.2075	100.5	0.281	0.001
16	1165	1029	15.3941	136.1807	101.5	1.342	0.010

*The SAS System**The REG Procedure**Model: MODEL1**Dependent Variable: price*

Output Statistics							
Obs	Dependent Variable	Predicted Value	Std Error Mean Predict	Residual	Std Error Residual	Student Residual	Cook's D
17	780	1078	24.4079	-298.0533	99.729	-2.989	0.134
18	800	1078	24.4079	-278.0533	99.729	-2.788	0.116
19	1180	1144	16.1917	35.5082	101.4	0.350	0.001
20	1000	1047	20.8165	-46.7925	100.5	-0.465	0.002
21	1200	1181	18.7001	18.7748	101.0	0.186	0.000
22	1000	1102	16.1357	-102.2859	101.4	-1.009	0.006
23	1025	1102	16.1357	-77.2859	101.4	-0.762	0.004
24	1175	1181	18.7001	-6.2252	101.0	-0.062	0.000
25	800	1029	15.3941	-228.8193	101.5	-2.254	0.029
26	915	1066	14.5648	-150.5526	101.6	-1.481	0.011
27	1185	973.3258	23.6612	211.6742	99.908	2.119	0.063
28	1020	967.8533	18.8522	52.1467	100.9	0.517	0.002
29	775	955.3526	22.4918	-180.3526	100.2	-1.800	0.041
30	850	997.5585	25.3530	-147.5585	99.493	-1.483	0.036

*The SAS System**The REG Procedure**Model: MODEL1**Dependent Variable: price*

Output Statistics							
Obs	Dependent Variable	Predicted Value	Std Error Mean Predict	Residual	Std Error Residual	Student Residual	Cook's D
31	1200	1139	19.5360	60.9807	100.8	0.605	0.003
32	1200	1102	16.1357	97.7141	101.4	0.964	0.006
33	775	918.6192	27.3849	-143.6192	98.953	-1.451	0.040
34	775	955.3526	22.4918	-180.3526	100.2	-1.800	0.041
35	1200	1084	21.9014	116.4742	100.3	1.161	0.016
36	1135	1066	14.5648	69.4474	101.6	0.683	0.002
37	1000	955.3526	22.4918	44.6474	100.2	0.446	0.003
38	1185	1084	21.9014	101.4742	100.3	1.012	0.012
39	1155	1102	16.1357	52.7141	101.4	0.520	0.002
40	1155	1181	18.7001	-26.2252	101.0	-0.260	0.001
41	1175	1181	18.7001	-6.2252	101.0	-0.062	0.000
42	1200	1181	18.7001	18.7748	101.0	0.186	0.000
43	1165	1102	16.1357	62.7141	101.4	0.619	0.002
44	1000	992.0859	18.3001	7.9141	101.0	0.078	0.000
45	1200	1120	24.4762	79.7408	99.712	0.800	0.010
46	1175	1078	24.4079	96.9467	99.729	0.972	0.014

*The SAS System**The REG Procedure**Model: MODEL1**Dependent Variable: price*

Output Statistics							
Obs	Dependent Variable	Predicted Value	Std Error Mean Predict	Residual	Std Error Residual	Student Residual	Cook's D
47	1000	967.8533	18.8522	32.1467	100.9	0.319	0.001
48	1125	1029	15.3941	96.1807	101.5	0.947	0.005
49	1150	1108	15.7390	42.2415	101.5	0.416	0.001
50	1125	1181	18.7001	-56.2252	101.0	-0.557	0.003
51	875	931.1199	20.5288	-56.1199	100.6	-0.558	0.003
52	1000	973.3258	23.6612	26.6742	99.908	0.267	0.001
53	1185	1041	21.0372	143.6801	100.5	1.430	0.022
54	1185	1066	14.5648	119.4474	101.6	1.175	0.007
55	1000	1066	14.5648	-65.5526	101.6	-0.645	0.002
56	950	1047	20.8165	-96.7925	100.5	-0.963	0.010
57	875	992.0859	18.3001	-117.0859	101.0	-1.159	0.011
58	775	931.1199	20.5288	-156.1199	100.6	-1.552	0.025
59	1100	1078	24.4079	21.9467	99.729	0.220	0.001
60	1125	1139	19.5360	-14.0193	100.8	-0.139	0.000
61	1200	1139	19.5360	60.9807	100.8	0.605	0.003
62	1175	1108	15.7390	67.2415	101.5	0.663	0.003

*The SAS System**The REG Procedure**Model: MODEL1**Dependent Variable: price*

Output Statistics							
Obs	Dependent Variable	Predicted Value	Std Error Mean Predict	Residual	Std Error Residual	Student Residual	Cook's D
63	1000	1047	20.8165	-46.7925	100.5	-0.465	0.002
64	1000	1084	21.9014	-83.5258	100.3	-0.833	0.008
65	1000	931.1199	20.5288	68.8801	100.6	0.685	0.005
66	1000	1034	20.9289	-34.2918	100.5	-0.341	0.001
67	985	967.8533	18.8522	17.1467	100.9	0.170	0.000
68	1150	1102	16.1357	47.7141	101.4	0.471	0.001
69	1150	1066	14.5648	84.4474	101.6	0.831	0.004
70	1075	955.3526	22.4918	119.6474	100.2	1.194	0.018
71	1050	931.1199	20.5288	118.8801	100.6	1.182	0.015
72	885	931.1199	20.5288	-46.1199	100.6	-0.458	0.002
73	1200	1108	15.7390	92.2415	101.5	0.909	0.005
74	1150	1102	16.1357	47.7141	101.4	0.471	0.001
75	1000	1005	19.0381	-4.5866	100.9	-0.045	0.000
76	1075	1078	24.4079	-3.0533	99.729	-0.031	0.000
77	980	1066	14.5648	-85.5526	101.6	-0.842	0.004
78	980	1102	16.1357	-122.2859	101.4	-1.206	0.009

*The SAS System**The REG Procedure**Model: MODEL1**Dependent Variable: price*

Output Statistics							
Obs	Dependent Variable	Predicted Value	Std Error Mean Predict	Residual	Std Error Residual	Student Residual	Cook's D
79	1000	1034	20.9289	-34.2918	100.5	-0.341	0.001
80	1085	1102	16.1357	-17.2859	101.4	-0.170	0.000
81	1175	1029	15.3941	146.1807	101.5	1.440	0.012
82	1150	1010	21.4489	139.9409	100.4	1.394	0.022
83	1175	1102	16.1357	72.7141	101.4	0.717	0.003
84	1000	1139	19.5360	-139.0193	100.8	-1.379	0.018
85	875	997.5585	25.3530	-122.5585	99.493	-1.232	0.025
86	995	1071	17.5024	-76.0251	101.2	-0.751	0.004
87	1000	1144	16.1917	-144.4918	101.4	-1.425	0.013
88	1250	1144	16.1917	105.5082	101.4	1.041	0.007
89	1150	1181	18.7001	-31.2252	101.0	-0.309	0.001
90	1150	1071	17.5024	78.9749	101.2	0.781	0.005
91	1200	1071	17.5024	128.9749	101.2	1.275	0.012
92	980	918.6192	27.3849	61.3808	98.953	0.620	0.007
93	995	1041	21.0372	-46.3199	100.5	-0.461	0.002

*The SAS System**The REG Procedure**Model: MODEL1**Dependent Variable: price*

Output Statistics							
Obs	Dependent Variable	Predicted Value	Std Error Mean Predict	Residual	Std Error Residual	Student Residual	Cook's D
94	1000	1144	16.1917	-144.4918	101.4	-1.425	0.013
95	1165	1144	16.1917	20.5082	101.4	0.202	0.000
96	1175	1029	15.3941	146.1807	101.5	1.440	0.012
97	1000	1034	20.9289	-34.2918	100.5	-0.341	0.001
98	775	1010	21.4489	-235.0591	100.4	-2.341	0.063
99	1000	1144	16.1917	-144.4918	101.4	-1.425	0.013
100	1015	1108	15.7390	-92.7585	101.5	-0.914	0.005
101	1200	1108	15.7390	92.2415	101.5	0.909	0.005
102	1175	1034	20.9289	140.7082	100.5	1.400	0.021
103	1095	1108	15.7390	-12.7585	101.5	-0.126	0.000
104	1100	1102	16.1357	-2.2859	101.4	-0.023	0.000
105	980	1041	21.0372	-61.3199	100.5	-0.610	0.004
106	1110	1010	21.4489	99.9409	100.4	0.995	0.011
107	1200	1181	18.7001	18.7748	101.0	0.186	0.000
108	1185	1108	15.7390	77.2415	101.5	0.761	0.003

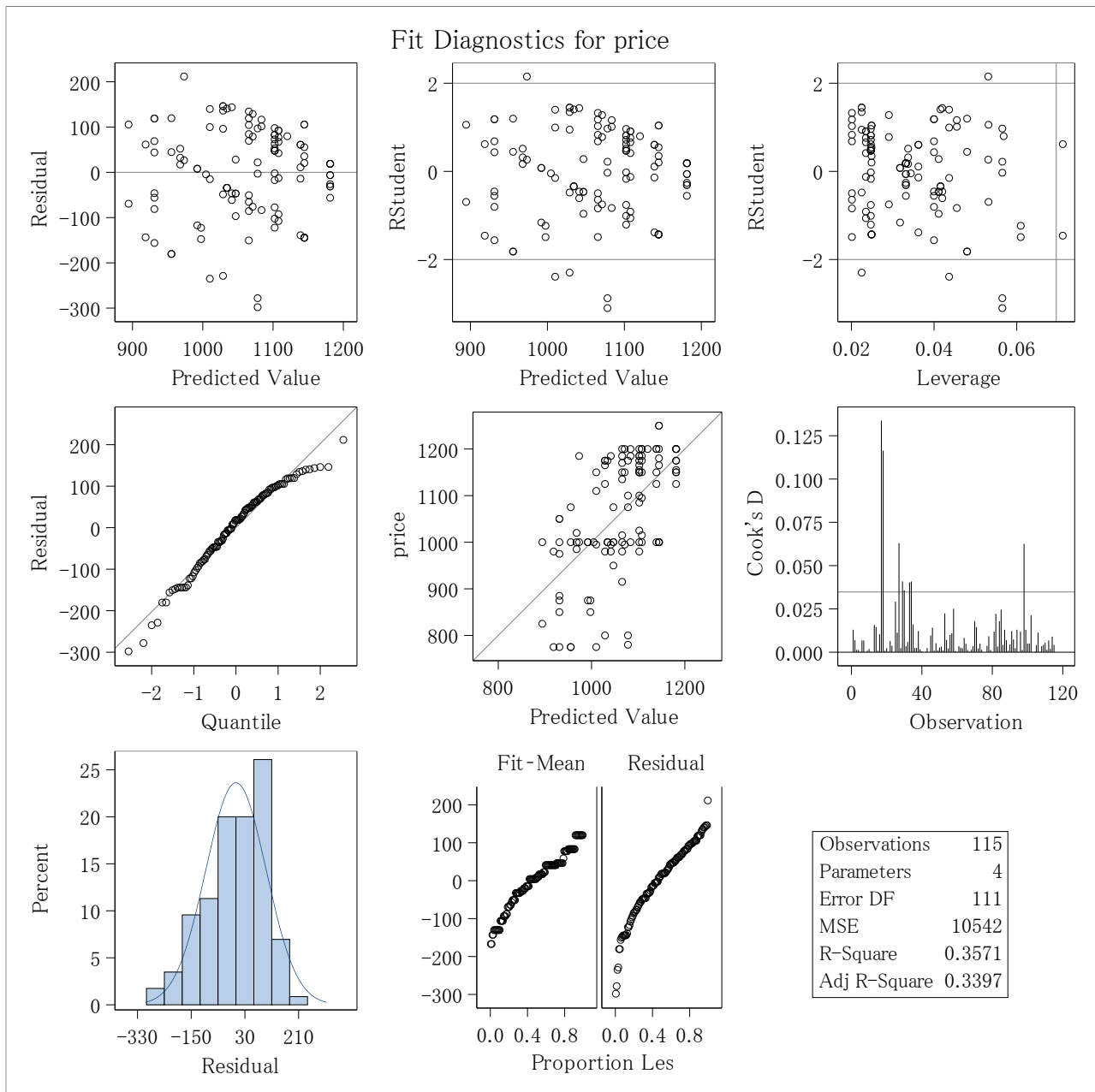
*The SAS System**The REG Procedure**Model: MODEL1**Dependent Variable: price*

Output Statistics							
Obs	Dependent Variable	Predicted Value	Std Error Mean Predict	Residual	Std Error Residual	Student Residual	Cook's D
109	1185	1102	16.1357	82.7141	101.4	0.816	0.004
110	1170	1066	14.5648	104.4474	101.6	1.028	0.005
111	1150	1102	16.1357	47.7141	101.4	0.471	0.001
112	1000	1108	15.7390	-107.7585	101.5	-1.062	0.007
113	975	931.1199	20.5288	43.8801	100.6	0.436	0.002
114	1200	1066	14.5648	134.4474	101.6	1.323	0.009
115	1185	1102	16.1357	82.7141	101.4	0.816	0.004

The SAS System

The REG Procedure

Model: MODEL1



Interpretation of plots of the final model.

Residual vs Predicted value: The points are randomly scattered around zero with no clear pattern which suggests that the assumptions of linearity is met and that there is evidence of constant variance.

RStudent vs Predicted value: It depicts no points outside three standard deviation from the mean therefore there is no outlier in the data.

Percent vs Residual: The histogram depicts a bell shaped (symmetric) curve which indicated that the errors are normally distributed.

MSE: The average variance of the residuals is estimated to be around 10542

Adjusted R-squared:

The model explains about 34% of the variation in price, after accounting for model complexity, indicating weak fit, with approximately 66% of the variation remaining unexplained.

*The SAS System**The REG Procedure**Model: MODEL1*

ODS RTF ;

Data cows;

input price age bred angus frame weight conditioned registered;

datalines;

1000	3	1	1	1	10.15	1	0
1250	3	1	1	1	11	1	0
980	5	1	0	0	11.15	1	0
1015	4	0	1	0	11	1	0
995	5	1	0	1	10	0	0
825	7	1	0	0	9.8	0	0
850	6	1	0	0	10.25	0	1
1150	2	1	1	0	10.5	1	1
1150	2	1	1	1	10.75	1	1
1200	3	1	1	1	11.75	1	0
1200	2	1	1	1	11.6	1	0
1000	6	1	1	0	11	1	0
1000	7	0	1	0	10	0	0
1050	6	1	0	0	10	0	1
1075	4	1	1	1	9.9	0	1
1165	5	1	1	0	11.35	1	0
780	2	0	1	0	8.85	0	0
800	2	0	1	0	9.5	0	1
1180	3	0	0	1	11.45	1	1
1000	4	0	0	1	11.45	0	1
1200	2	1	0	1	11.5	1	1
1000	3	1	1	0	10	1	0
1025	3	1	1	0	9.75	1	0
1175	2	0	1	1	11.35	1	1
800	5	1	0	0	12	1	0
915	4	0	1	0	11.85	1	1
1185	6	1	1	1	12	0	0
1020	5	1	1	0	11.25	0	0
775	7	1	1	0	12	1	0
850	7	1	0	1	12	1	1
1200	2	1	1	0	10	1	1
1200	3	1	1	0	10.15	1	1
775	8	1	0	0	12.5	1	0
775	7	1	0	0	11.85	1	0
1200	3	1	0	1	11.85	0	1
1135	4	1	0	0	12	1	0
1000	7	1	0	0	11.5	1	0
1185	3	0	0	1	12	0	1
1155	3	1	0	0	11.85	1	0
1155	2	0	0	1	12	1	1
1175	2	1	1	1	11.75	1	0
1200	2	1	0	1	11.5	1	0
1165	3	1	1	0	11.65	1	0
1000	6	1	1	0	12.25	1	0
1200	2	0	1	1	10.25	0	0
1175	2	1	0	0	10.25	0	1

*The SAS System**The REG Procedure**Model: MODEL1*

1000	5	0	0	0	10	0	1
1125	5	0	0	0	10	1	0
1150	4	1	1	1	12.25	1	0
1125	2	0	0	1	11.25	1	1
875	6	0	0	0	10	0	0
1000	6	1	0	1	11.25	0	0
1185	3	1	1	0	10	0	0
1185	4	1	1	0	10	1	1
1000	4	1	1	0	11	1	0
950	4	1	1	1	10.25	0	0
875	6	1	0	0	11.25	1	0
775	6	0	0	0	10.15	0	0
1100	2	0	1	0	9.85	0	1
1125	2	0	1	0	10.35	1	1
1200	2	0	0	0	10.65	1	1
1175	4	1	1	1	12.3	1	0
1000	4	1	0	1	11.75	0	0
1000	3	1	0	1	11.85	0	0
1000	6	1	1	0	11.25	0	0
1000	6	1	0	1	11.25	1	0
985	5	0	1	0	10.5	0	0
1150	3	1	1	0	11.75	1	1
1150	4	1	1	0	11.5	1	0
1075	7	1	0	0	11.45	1	0
1050	6	0	0	0	11.5	0	0
885	6	0	0	0	10	0	0
1200	4	1	1	1	11.5	1	1
1150	3	1	1	0	11.5	1	1
1000	4	1	0	0	10.85	0	0
1075	2	1	1	0	10.85	0	1
980	4	0	0	0	12.5	1	0
980	3	0	0	0	12.45	1	0
1000	6	1	1	1	12	1	0
1085	3	1	0	0	11.25	1	0
1175	5	1	0	0	11.55	1	0
1150	5	1	0	1	11.15	0	1
1175	3	1	0	0	11.15	1	0
1000	2	0	1	0	10	1	0
875	7	1	1	1	11.65	1	0
995	5	1	1	1	11.55	1	1
1000	3	1	1	1	10.15	1	1
1250	3	1	1	1	11	1	1
1150	2	1	0	1	11	1	0
1150	5	1	0	1	10.85	1	0
1200	5	1	0	1	10.95	1	1
980	8	1	1	0	12.75	1	0
995	3	0	0	0	10	0	0
1000	3	1	1	1	10.35	1	0
1165	3	1	0	1	10.35	1	0
1175	5	0	1	0	11.25	1	1

*The SAS System**The REG Procedure**Model: MODEL1*

1000	6	1	1	1	11.2	1	0
775	5	0	1	1	10	0	0
1000	3	1	1	1	11.65	1	0
1015	4	1	0	1	11.5	1	0
1200	4	1	1	1	11.5	1	0
1175	6	1	1	1	11.75	1	1
1095	4	0	1	1	11.75	1	0
1100	3	1	1	0	12.35	1	1
980	3	0	0	0	10	0	0
1110	5	0	1	1	10.35	0	0
1200	2	1	1	1	11	1	0
1185	4	0	1	1	11.25	1	0
1185	3	0	1	0	11	1	0
1170	4	0	1	0	11.25	1	0
1150	3	1	0	0	11.1	1	0
1000	4	1	1	1	11	1	1
975	6	1	0	0	10	0	0
1200	4	0	1	0	11	1	0
1185	3	1	1	0	11	1	0

;

RUN;**PROC CORR** Data = COWS;

var price age bred angus frame weight conditioned registered;

RUN;**PROC REG DATA=** COWS **ALPHA= 0.05**;**MODEL** price= age bred angus frame weight conditioned registered;**RUN;****PROC REG DATA=** COWS **ALPHA= 0.05**;**MODEL** price= age bred angus frame weight conditioned registered/ **CLB R VIF SELECTION=**
BACKWARD SLS=0.05 LACKFIT DW ;**RUN;****ODS RTF CLOSE;**